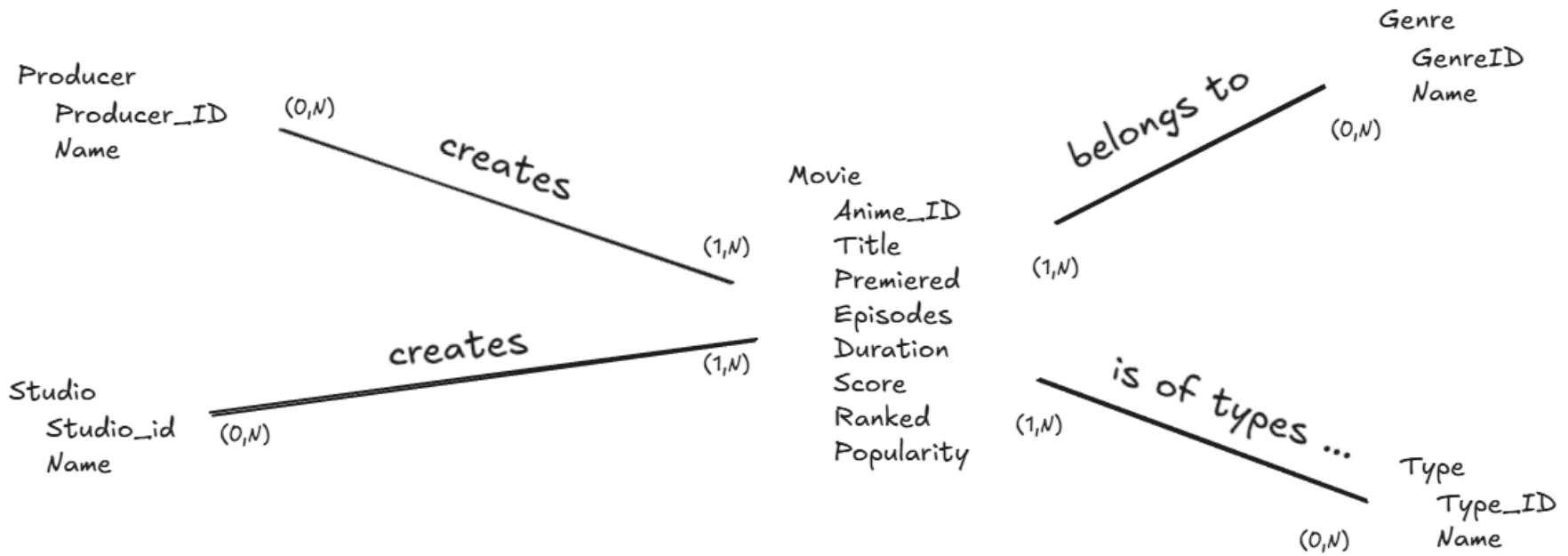
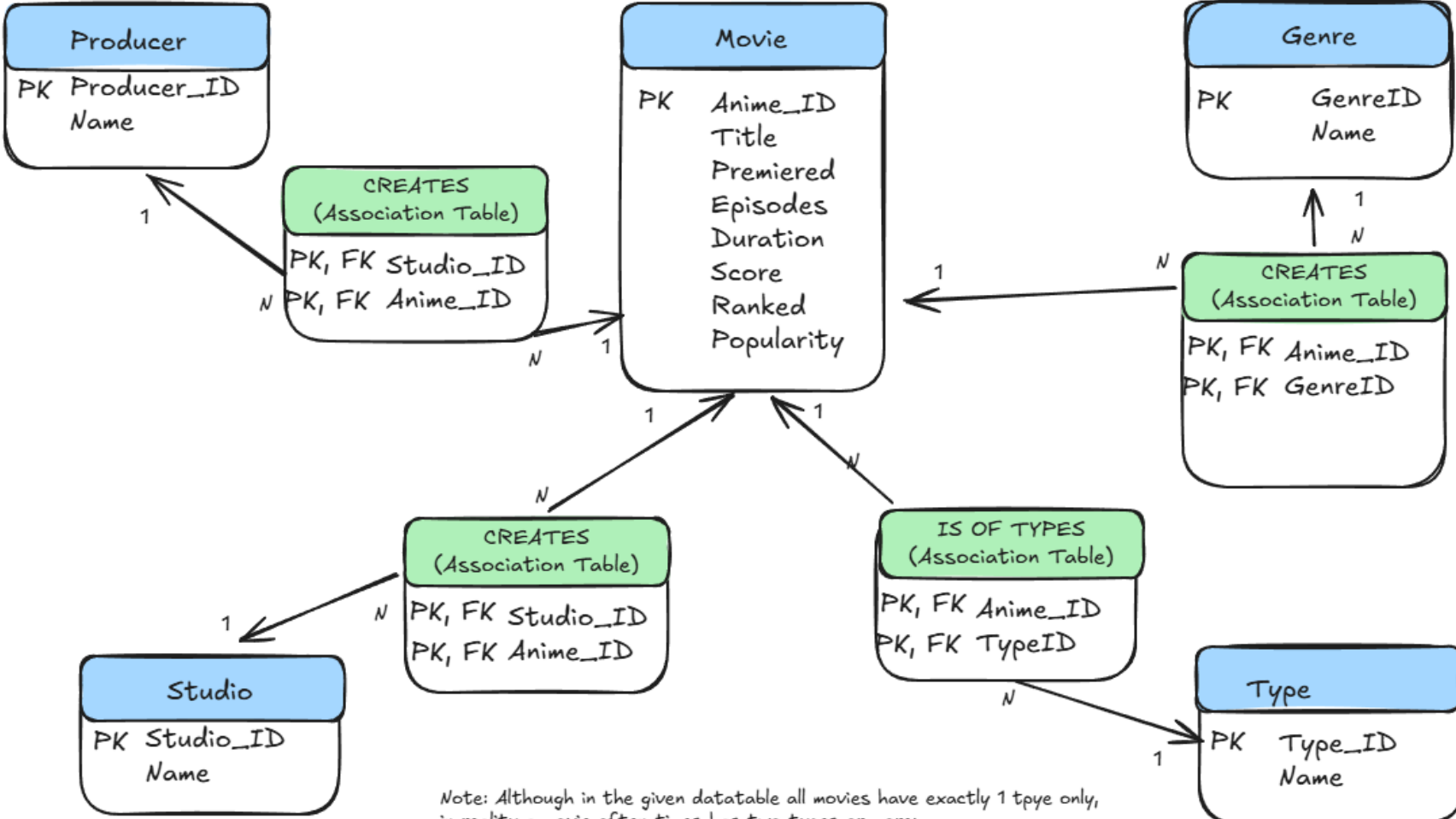


MCD



⇒ MLD See next page

MLD



Note: Although in the given datatable all movies have exactly 1 type only, in reality a movie often times has two types or more: Movie and then shown on tv. Thus Type is an own entity.

⇒ SQL Queries see next page

Implementation

```
-- =====  
-- PART 1A: CREATE TABLES AND RELATIONSHIPS  
-- =====  
  
BEGIN;  
  
CREATE TABLE type (  
    type_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    name VARCHAR(30) NOT NULL UNIQUE  
);  
  
CREATE TABLE source (  
    source_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    name VARCHAR(50) NOT NULL UNIQUE  
);  
  
CREATE TABLE genre (  
    genre_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    name VARCHAR(50) NOT NULL UNIQUE  
);  
  
CREATE TABLE producer (  
    producer_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    name VARCHAR(100) NOT NULL UNIQUE  
);  
  
CREATE TABLE studio (  
    studio_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    name VARCHAR(100) NOT NULL UNIQUE  
);  
  
CREATE TABLE anime (  
    anime_id INTEGER PRIMARY KEY,  
    english_name VARCHAR(255) NOT NULL,  
    score NUMERIC(4,2),
```

```
episodes INTEGER,  
aired VARCHAR(100),  
premiered VARCHAR(30),  
duration VARCHAR(50),  
rating VARCHAR(100),  
ranked INTEGER,  
popularity INTEGER,  
type_id INTEGER NOT NULL,  
source_id INTEGER NOT NULL,
```

```
CONSTRAINT fk_anime_type  
    FOREIGN KEY (type_id)  
    REFERENCES type(type_id),
```

```
CONSTRAINT fk_anime_source  
    FOREIGN KEY (source_id)  
    REFERENCES source(source_id)
```

```
);
```

```
CREATE TABLE anime_genre (  
    anime_id INTEGER NOT NULL,  
    genre_id INTEGER NOT NULL,  
  
    PRIMARY KEY (anime_id, genre_id),  
  
    FOREIGN KEY (anime_id)  
        REFERENCES anime(anime_id),  
  
    FOREIGN KEY (genre_id)  
        REFERENCES genre(genre_id)  
);
```

```
CREATE TABLE anime_producer (  
    anime_id INTEGER NOT NULL,  
    producer_id INTEGER NOT NULL,  
  
    PRIMARY KEY (anime_id, producer_id),
```

```
FOREIGN KEY (anime_id)
  REFERENCES anime(anime_id),

FOREIGN KEY (producer_id)
  REFERENCES producer(producer_id)
);
```

```
CREATE TABLE anime_studio (
  anime_id INTEGER NOT NULL,
  studio_id INTEGER NOT NULL,

  PRIMARY KEY (anime_id, studio_id),

  FOREIGN KEY (anime_id)
    REFERENCES anime(anime_id),

  FOREIGN KEY (studio_id)
    REFERENCES studio(studio_id)
);
```

```
COMMIT;
```

```
-- =====
-- PART 1B: INSERT DATA
-- =====
```

```
BEGIN;
```

```
INSERT INTO type (name)
VALUES
  ('TV'),
  ('Movie');
```

```
INSERT INTO source (name)
VALUES
  ('Original'),
  ('Manga');
```

```
INSERT INTO genre (name)
```

```
VALUES
```

```
  ('Action'),  
  ('Adventure'),  
  ('Drama'),  
  ('Sci-Fi'),  
  ('Space'),  
  ('Shounen'),  
  ('Military');
```

```
INSERT INTO producer (name)
```

```
VALUES
```

```
  ('Bandai Visual'),  
  ('Sunrise'),  
  ('TV Tokyo'),  
  ('Shueisha'),  
  ('Fuji TV'),  
  ('Sotsu'),  
  ('Sony Music Entertainment');
```

```
INSERT INTO studio (name)
```

```
VALUES
```

```
  ('Sunrise'),  
  ('Bones'),  
  ('Studio Pierrot'),  
  ('Toei Animation');
```

```
-- =====  
-- INSERT ANIME  
-- =====
```

```
INSERT INTO anime (
```

```
  anime_id,  
  english_name,  
  score,  
  episodes,
```

```

aired,
premiered,
duration,
rating,
ranked,
popularity,
type_id,
source_id
)
VALUES
(
  1,
  'Cowboy Bebop',
  8.78,
  26,
  'Apr 3, 1998 to Apr 24, 1999',
  'Spring 1998',
  '24 min. per ep',
  'R - 17+ (violence & profanity)',
  28,
  39,
  (SELECT type_id FROM type WHERE name = 'TV'),
  (SELECT source_id FROM source WHERE name = 'Original')
),
(
  2,
  'Cowboy Bebop:The Movie',
  8.39,
  1,
  'Sep 1, 2001',
  NULL,
  '1 hr. 55 min.',
  'R - 17+ (violence & profanity)',
  159,
  518,
  (SELECT type_id FROM type WHERE name = 'Movie'),
  (SELECT source_id FROM source WHERE name = 'Original')
),

```

```

(
  3,
  'Naruto',
  7.91,
  220,
  'Oct 3, 2002 to Feb 8, 2007',
  'Fall 2002',
  '23 min. per ep.',
  'PG-13 - Teens 13 or older',
  660,
  8,
  (SELECT type_id FROM type WHERE name = 'TV'),
  (SELECT source_id FROM source WHERE name = 'Manga')
),
(
  4,
  'One Piece',
  8.52,
  NULL,
  'Oct 20, 1999 to ?',
  'Fall 1999',
  '24 min.',
  'PG-13 - Teens 13 or older',
  95,
  31,
  (SELECT type_id FROM type WHERE name = 'TV'),
  (SELECT source_id FROM source WHERE name = 'Manga')
),
(
  5,
  'Mobile Suit Gundam SEED',
  7.79,
  50,
  'Oct 5, 2002 to Sep 27, 2003',
  'Fall 2002',
  '24 min. per ep.',
  'R - 17+ (violence & profanity)',
  850,

```



```

1057,
(SELECT type_id FROM type WHERE name = 'TV'),
(SELECT source_id FROM source WHERE name = 'Original')
),
(
6,
'Mobile Suit Gundam SEED Destiny',
7.22,
50,
'Oct 9, 2004 to Oct 1, 2005',
'Fall 2004',
'24 min. per ep.',
'R - 17+ (violence & profanity)',
2687,
1530,
(SELECT type_id FROM type WHERE name = 'TV'),
(SELECT source_id FROM source WHERE name = 'Original')
);

```

```

INSERT INTO anime_studio (anime_id, studio_id)
SELECT
    v.anime_id,
    s.studio_id
FROM (
    VALUES
        (1, 'Sunrise'),
        (2, 'Bones'),
        (3, 'Studio Pierrot'),
        (4, 'Toei Animation'),
        (5, 'Sunrise'),
        (6, 'Sunrise')
) AS v(anime_id, studio_name)
JOIN studio s
    ON s.name = v.studio_name;

```

```

INSERT INTO anime_producer (anime_id, producer_id)
SELECT

```

```

v.anime_id,
p.producer_id
FROM (
VALUES
  (1, 'Bandai Visual'),
  (2, 'Sunrise'),
  (2, 'Bandai Visual'),
  (3, 'TV Tokyo'),
  (3, 'Shueisha'),
  (4, 'Fuji TV'),
  (4, 'Shueisha'),
  (5, 'Sotsu'),
  (5, 'Sony Music Entertainment'),
  (6, 'Sotsu'),
  (6, 'Sony Music Entertainment')
) AS v(anime_id, producer_name)
JOIN producer p
  ON p.name = v.producer_name;

```

```

INSERT INTO anime_genre (anime_id, genre_id)
SELECT
  v.anime_id,
  g.genre_id
FROM (
VALUES
  (1, 'Action'),
  (1, 'Adventure'),
  (1, 'Drama'),
  (1, 'Sci-Fi'),
  (1, 'Space'),

  (2, 'Action'),
  (2, 'Drama'),
  (2, 'Sci-Fi'),
  (2, 'Space'),

  (3, 'Action'),
  (3, 'Adventure'),

```

(3, 'Shounen'),

(4, 'Action'),

(4, 'Adventure'),

(4, 'Shounen'),

(5, 'Action'),

(5, 'Drama'),

(5, 'Military'),

(5, 'Sci-Fi'),

(5, 'Space'),

(6, 'Action'),

(6, 'Drama'),

(6, 'Military'),

(6, 'Sci-Fi'),

(6, 'Space')

) AS v(anime_id, genre_name)

JOIN genre g

ON g.name = v.genre_name;